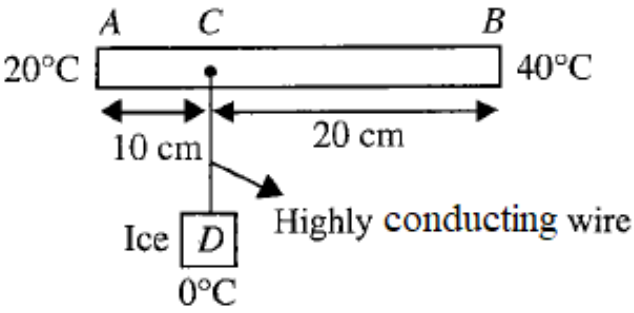
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

As shown in Fig. AB is a rod of length $30\ \text{cm}$ and area of cross section $1.0\ \text{cm}^2$ and thermal conductivity 336 SI units. The ends A and B are maintained at temperatures 20°C and 40°C, respectively. A point C of this rod is connected to a box D, containing ice at 0°C, through a highly conducting wire of negligible heat capacity. The rate at which ice melts in the box is (assume latent heat of fusion for ice $L_f = 80\text{cal/g}$)



(A) 84mg/s

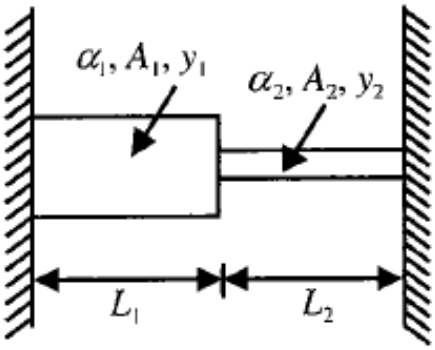
(B) $84\ \text{g/s}$

(C) 20mg/s

(D) 40mg/s

Q.2

Two elastic rods are joined between fixed supports as shown in Fig. Condition for no change in the lengths of individual rods with the increase of temperature ($\alpha_1, \alpha_2 =$ linear expansion coefficient, $A_1, A_2 =$ area of rods, $y_1, y_2 =$ Young's modulus) is



(A) $\frac{A_1}{A_2} = \frac{\alpha_1 y_1}{\alpha_2 y_2}$

(B) $\frac{A_1}{A_2} = \frac{L_1 \alpha_1 y_1}{L_2 \alpha_2 y_2}$

(C) $\frac{A_1}{A_2} = \frac{L_2 \alpha_2 y_2}{L_1 \alpha_1 y_1}$

(D) $\frac{A_1}{A_1} = \frac{\alpha_2 y_2}{\alpha_1 y_1}$

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Q.3

Two rods of length L_2 and coefficient of linear expansion α_2 are connected freely to a third rod of length L_1 of coefficient of linear expansion α_1 to form an isosceles triangle. The arrangement is supported on the knife edge at the midpoint of L_1 which is horizontal. The apex of the isosceles triangle is to remain at a constant distance from the knife edge if

(A) $\frac{L_1}{L_2} = \frac{\alpha_2}{\alpha_1}$

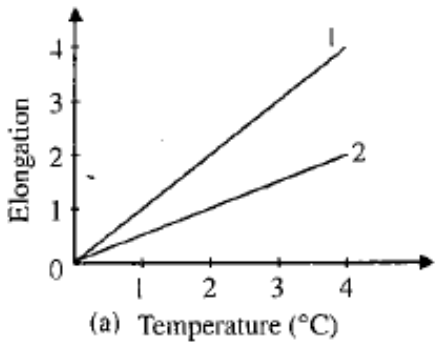
(B) $\frac{L_1}{L_2} = \sqrt{\frac{\alpha_2}{\alpha_1}}$

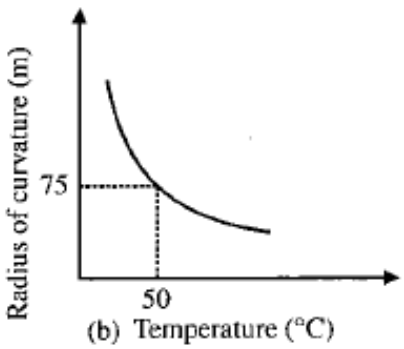
(C) $\frac{L_1}{L_2} = 2\frac{\alpha_2}{\alpha_1}$

(D) $\frac{L_1}{L_2} = 2\sqrt{\frac{\alpha_2}{\alpha_1}}$

Q.4

Figure shows the graphs of elongation versus temperature for two different metals. If these metals are employed to form a straight bimetallic strip of thickness $6\ \text{cm}$ and heated, it bends in the form of an arc, the radius of curvature changing with temperature approximately as shown in the figure. The linear expansivities of the two metals are





(A) $24 \times 10^{-6}/^{\circ}\text{C}$ and $12 \times 10^{-6}/^{\circ}\text{C}$

(B) $20 \times 10^{-6}/^{\circ}\text{C}$ and $10 \times 10^{-6}/^{\circ}\text{C}$

(C) $18 \times 10^{-6/}{}^{\circ}\text{C}$ and $9 \times 10^{-6}/^{\circ}\text{C}$

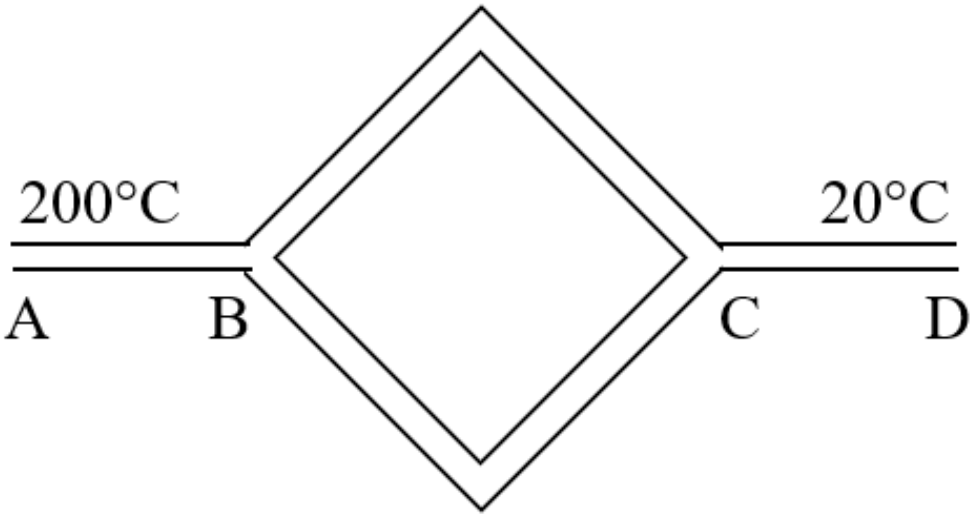
(D) $16 \times 10^{-6/}{}^{\circ}\text{C}$ and $8 \times 10^{-6}/^{\circ}\text{C}$

Q.5

Six identical conducting rods are joined as shown in Fig. Points A and D are maintained at temperatures

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$200^\circ C$ and $20^\circ C$, respectively. The temperature of junction B will be



(A) $120^\circ C$

(B) $100^\circ C$

(C) $140^\circ C$

(D) $80^\circ C$

Q.6

A solid copper cube of edges $1\ \text{cm}$ is suspended in an evacuated enclosure. Its temperature is found to fall from $100^\circ C$ to $99^\circ C$ in $100\ \text{s}$. Another solid copper cube of edges $2\ \text{cm}$, with similar surface nature, is suspended in a similar manner. The time required for this cube to cool from $100^\circ C$ to $99^\circ C$ will be approximately

(A) $25\ \text{s}$

(B) $50\ \text{s}$

(C) $200\ \text{s}$

(D) $400\ \text{s}$

Q.7

A sphere and a cube of same material and same volume are heated up to same temperature and allowed to cool in the same surroundings. The ratio of the amounts of radiations emitted in equal time intervals will be

(A) $1:1$

(B) $\frac{4\pi}{3}:1$

(C) $\left(\frac{\pi}{6}\right)^{1/3}:1$

(D) $\frac{1}{2}\left(\frac{4\pi}{3}\right)^{2/3}:1$

Q.8

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A liquid of mass m and specific heat c is heated to a temperature $2T$. Another liquid of mass $m/2$ and specific heat $2c$ is heated to a temperature T. If these two liquids are mixed, the resulting temperature of the mixture is

(A) $(2/3)T$

(B) $(8/5)T$

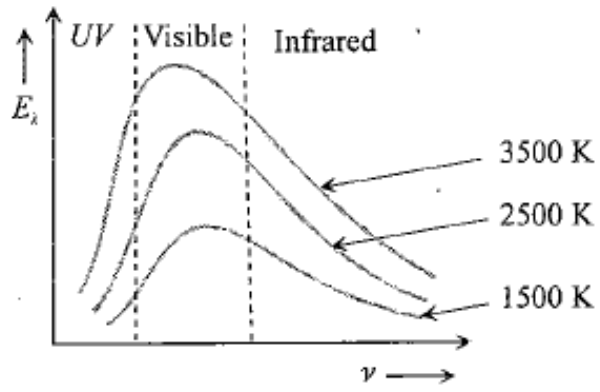
(C) $(3/5)T$

(D) $(3/2)T$

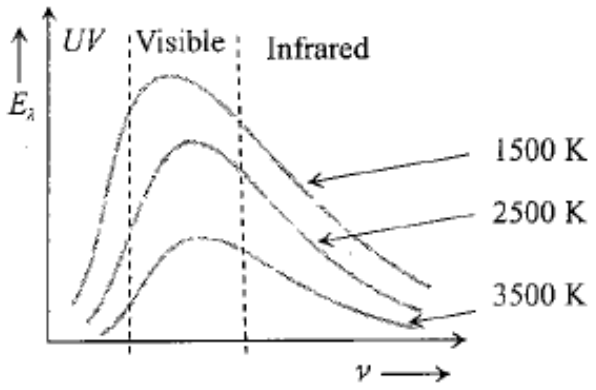
Q.9

Following graph shows the correct variation in intensity of heat radiations by black body and frequency at a fixed temperature.

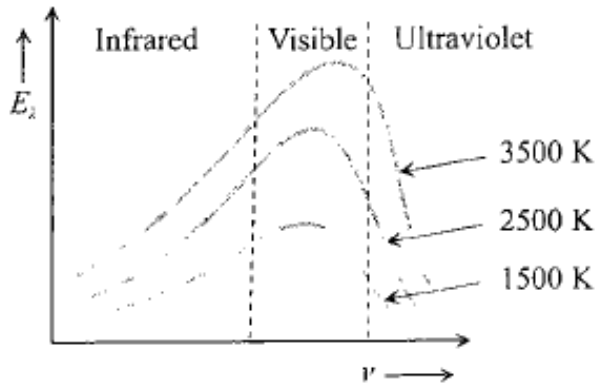
(A)



(B)

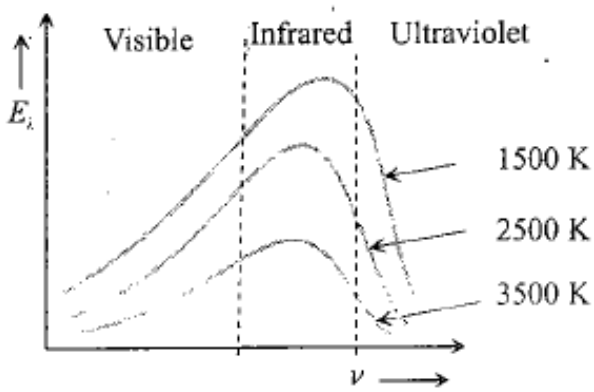


(C)



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(D)



Q.10

The coefficient of apparent expansion of a liquid in a copper vessel is C and in a silver vessel S. The coefficient of volume expansion of copper is γ_C. What is the coefficient of linear expansion of silver

(A) $(C + \gamma_C + S)/3$

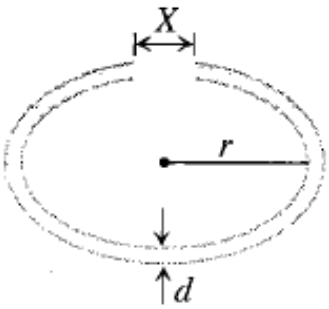
(B) $(C - \gamma_c + S)/3$

(C) $(C + \gamma_C - S)/3$

(D) $(C - \gamma_C - S)/3$

Q.11

A cylindrical metal rod of length L_0 is shaped into a ring with a small gap as shown. On heating the system



(A) x decreases, r and d increase

(B) x and r increase, d decreases

(C) x, r and d all increase

(D) Data insufficient to arrive at a conclusion

Q.12

Three liquids with masses m_1, m_2, m_3 are thoroughly mixed. If their specific heats are c_1, c_2, c_3 and their temperatures T_1, T_2, T_3, respectively, then the temperature of the mixture is

(A) $\frac{c_1T_1+c_2T_2+c_3T_3}{m_1c_1+m_2c_2+m_3c_3}$

(B) $\frac{m_1c_1T_1+m_2c_2T_2+m_3c_3T_3}{m_1c_1+m_2c_2+m_3c_3}$

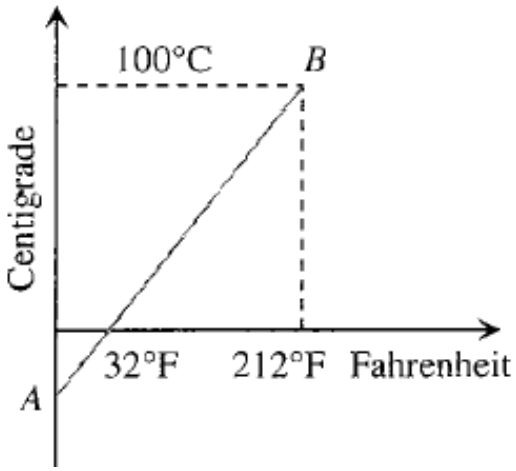
(C) $\frac{m_1c_1T_1+m_2c_2T_2+m_3c_3T_3}{m_1T_1+m_2T_2+m_3T_3}$

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(D) $\frac{m_1T_1+m_2T_2+m_3T_3}{c_1T_1+c_2T_2+c_3T_3}$

Q.13

The graph AB shown in Fig. is a plot of temperature of a body in degree celsius and degree Fahrenheit. Then



(A) Slope of line AB is $9/5$

(B) Slope of line AB is $5/9$

(C) Slope of line AB is $1/9$

(D) Slope of line AB is $3/9$

Q.14

Four spheres A, B, C and D have their radii in arithmetic progression and the specific heat capacities of their substances are in geometric progression. If the ratios of heat capacities of D and B to that of C and A are as $8:27$. The ratio of masses of B and A is (assume same density for all spheres)

(A) $8:1$

(B) $4:1$

(C) $1:8$

(D) $1:4$

Q.15

A liquid takes 5 min to cool from 80°C to 50°C. How much time will it take to cool from 60°C to 30°C ? The temperature of the surrounding is 20°C.

(A) 5 min

(B) 9 min

(C) 4 min

(D) 12 min

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Q.16

Compared to a burn due to water at $100^{\circ}\mathrm{C}$; a burn due to steam at $100^{\circ}\mathrm{C}$ is

(A) More dangerous

(B) Less dangerous

(C) Equally dangerous

(D) None of these

Q.17

The energy spectrum of a black body exhibits a maximum around a wavelength λ_0. The temperature of the black body is now changed such that the energy is maximum around a wavelength $3\lambda_0/4$. The power radiated by the black body will now increase by a factor of

(A) $256/81$

(B) $64/27$

(C) $16/9$

(D) $4/3$

Q.18

A black body emits radiation at the rate P when its temperature is T. At this temperature the wavelength at which the radiation has maximum intensity is λ_0. If at another temperature T' the power radiated is P' and wavelength at maximum intensity is $\lambda_0/2$ then

(A) $P'T' = 32PT$

(B) $P'T' = 16PT$

(C) $P'T' = 8PT$

(D) $P'T' = 4PT$

Q.19

Heat is required to change $1\ \mathrm{kg}$ of ice at $-20^{\circ}\mathrm{C}$ into steam. Q_1 is the heat needed to warm the ice from $-20^{\circ}\mathrm{C}$ to $0^{\circ}\mathrm{C}$, Q_2 is the heat needed to melt the ice, Q_3 is the heat needed to warm the water from $0^{\circ}\mathrm{C}$ to $100^{\circ}\mathrm{C}$ and Q_4 is the heat needed to vapourize the water. Then

(A) $Q_4 > Q_3 > Q_2 > Q_1$

(B) $Q_{+} > Q_3 > Q_1 > Q$

(C) $Q_4 > Q_2 > Q_3 > Q_1$

(D) $Q_4 > Q_2 > Q_1 > Q_3$

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Q.20

The specific heat of a substance varies with temperature $t\ (^\circ\mathrm{C})$ as $c = 0.20 + 0.14t + 0.023t^2\ (\mathrm{cal/g/^\circ C})$ The heat required to raise the temperature of $2\ \mathrm{g}$ of substance from $5^\circ\mathrm{C}$ to $15^\circ\mathrm{C}$ will be

(A) 24 cal

(B) 56 cal

(C) 82 cal

(D) $100\mathrm{c}$

Q.21

Four cylindrical rods of same material with length and radius $(\ell, r), (2\ell, r), (2\ell, 2r)$ and $(\ell, 2r)$ are connected between two reservoirs at $0^\circ\mathrm{C}$ and $100^\circ\mathrm{C}$. Find the ratio of the maximum to minimum rate of conduction in them.

Q.22

An iron tyre is to be fitted onto a wooden wheel $1.0\ \mathrm{m}$ in diameter. The diameter of the tyre is $6\ \mathrm{mm}$ smaller than that of wheel. The tyre should be heated so that its temperature increases by a minimum of (coefficient of volume expansion of iron is $3.6 \times 10^{-5}/^\circ\mathrm{C}$)

Q.23

In two experiments with a continuous flow calorimeter to determine the specific heat capacity of a liquid, an input power of $16\ \mathrm{W}$ produced a rise of $10\ \mathrm{K}$ in the liquid. When the power was doubled, the same temperature rise was achieved by making the rate of flow of liquid three times faster. Find the power lost (in W) to the surrounding in each case.

Q.24

A body is cooled in 2 min in a room at temperature of $30^\circ\mathrm{C}$ from $75^\circ\mathrm{C}$ to $65^\circ\mathrm{C}$. If the same body is cooled from $55^\circ\mathrm{C}$ to $45^\circ\mathrm{C}$ in the same room, find the time taken (in minute).

Q.25

Two identical conducting rods are first, connected independently to two vessels, one containing water at $100^\circ\mathrm{C}$ and the other containing ice at $0^\circ\mathrm{C}$. In the second case, rods are joined end to end and are connected to the same vessels. If q_1 and q_2 (in $\mathrm{g/s}$) are the rates of melting of ice in two cases, then find the ratio of q_1/q_2.

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Q.26

Two vessels connected at the bottom by a thin pipe with a sliding plug contain liquid at $20^{\circ}\mathrm{C}$ and $80^{\circ}\mathrm{C}$ respectively. The coefficient of cubic expansion of liquid is $10^{-3}\ \mathrm{K}^{-1}$. The ratio of heights of liquid columns in the vessel $\left(\mathrm{H}_{2U} / \ \mathrm{H}_{\mathrm{B0}}\ \right)$ is nearest to which integer?

Q.27

$2\ \mathrm{kg}$ of ice at $-15^{\circ}\mathrm{C}$ is mixed with $2.5\ \mathrm{kg}$ of water at $25^{\circ}\mathrm{C}$ in an insulating container. If the specific heat capacities of ice and water are $0.5\mathrm{cal/g^{\circ}C}$ and $\mathrm{1cal/g^{\circ}C}$, find the amount of water present in the container? (in kg nearest integer)

Q.28

$2\ \mathrm{kg}$ of ice at $-20^{\circ}\mathrm{C}$ is mixed with $5\ \mathrm{kg}$ of water at $20^{\circ}\mathrm{C}$ in an insulating vessel having a negligible heat capacity. Calculate the final mass of water (in kg) remaining in the container.

Q.29

A clock with a metallic pendulum at $15^{\circ}\mathrm{C}$ runs faster by 5 s each day and at $30^{\circ}\mathrm{C}$, runs slow by $10\ \mathrm{s}$. Find the coefficient of linear expansion of the metal. (nearly in $\left(0^{-6}/^{\circ}\mathrm{C}\right)$

Q.30

A brass rod and a lead rod each $80\ \mathrm{cm}$ long at $0^{\circ}\mathrm{C}$ are clamped together at one end with their free ends coinciding. The separation of free ends of the rods if the system is placed in a steam bath is $\left(\alpha_{\text{brass}} = 18 \times 10^{-6}/^{\circ}\mathrm{C}\right.$ and $\left.\alpha_{\text{lead}} = 28 \times 10^{-6}/^{\circ}\mathrm{C}\right)$ to the nearest value (in mm) is

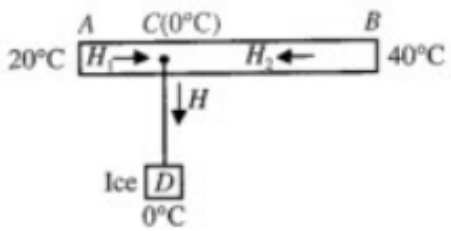
Answers & Solutions

Q.1 Answer:

$40\mathrm{mg/s}$

Solution:

Thermal resistance of $AC = \frac{L}{KA} = \frac{0.1}{336\times10^{-4}} = \frac{10}{336} = R$ **(let)**



Thermal resistance of $BC = \frac{0.2}{336\times10^{-4}} = 2R$

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Heat flow rates are

$$H_1 = \frac{20}{R}; H_2 = \frac{40}{2R} = \frac{20}{R}$$

$$H = H_1 + H_2 = \frac{40}{R} = \frac{40 \times 336}{10^5}$$

$$= \frac{13440}{10^3} = 13.44 \text{ W}$$

Q.2 Answer:

$\frac{A_1}{A_1} = \frac{\alpha_2 y_2}{\alpha_1 y_1}$

Solution:

Since tension in the two rods will be same,

$$A_1 Y_1 \alpha_1 \Delta\theta = A_2 Y_2 \alpha_2 \Delta\theta$$

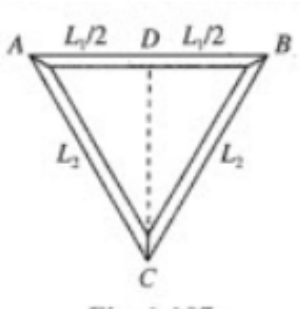
$$A_1 Y_1 \alpha_1 = A_2 Y_2 \alpha_2$$

Q.3 Answer:

$\frac{L_1}{L_2} = 2\sqrt{\frac{\alpha_2}{\alpha_1}}$

Solution:

The apex of the isosceles triangle to remain at a constant distance from the knife edge. Thus, DC should remain constant before and after heating.



Before expansion: In triangle ADC

$$(DC)^2 = L_2^2 - \left(\frac{L_1}{2}\right)^2$$

After expansion:

$$(DC)^2 = \left[L_2\left(1 + \alpha_2 t\right)\right]^2 - \left[\frac{L_1}{2}\left(1 + \alpha_1 t\right)\right]^2$$

Equating Eqs. (i) and (ii), we get

$$L_2^2 - \left(\frac{L_1}{2}\right)^2 = \left[L_2\left(1 + \alpha_2 t\right)\right]^2 - \left[\frac{L_1}{2}\left(1 + \alpha_1 t\right)\right]$$

Q.4 Answer:

$16 \times 10^{-6/\circ}\text{C}$ and $8 \times 10^{-6}/^\circ\text{C}$

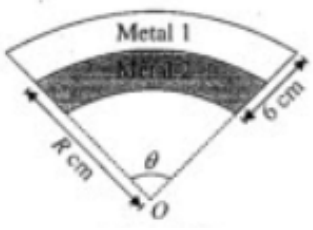
Solution:

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The slope of lines are,

$\left(\frac{\Delta L}{\Delta T}\right)_1 = L\alpha_1 = \frac{4-0}{4-0} = 1$ **(For the first metal)**

and $\left(\frac{\Delta L}{\Delta T}\right)_2 = L\alpha_2 = \frac{2-0}{4-0} = \frac{1}{2}$ **(For the second metal)**



$$L_0\left(1+\alpha_1 T\right) = (R+1.5)\theta$$

$$\text{and } L_0\left(1+\alpha_2 T\right) = (R-1.5)\theta$$

$\therefore$ **Dividing, we get,** $\frac{1+\alpha_1 T}{1+\alpha_2 T} = \frac{R+1.5}{R-1.5}$

or, $(\alpha_1 - \alpha_2)\,T = \frac{3}{R}$ **and** $\alpha_1 - \alpha_2 = \frac{3}{RT}$

From the graph [shown in Fig. 1.87(a)]

$\alpha_1 - \alpha_2 = \frac{3}{7500\times 50} = 8\times 10^{-6}/^{\circ}\mathrm{C}$

But $\alpha_1/\alpha_2 = 2$

$\therefore \quad a_1 = 16\times 10^{-6}/^{\circ}\mathrm{C}$ **and** $\quad a_2 = 8\times 10^{-6}/^{\circ}\mathrm{C}$

Q.5 Answer:

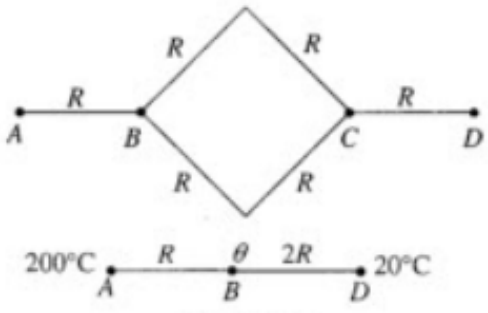
$140^{\circ}\mathrm{C}$

Solution:

Let the thermal resistance of each rod be R**.**

The two resistances connected along two paths from B **to\}** C **are equivalent to** $2R$ **each and their parallel combination is** R**.**

Effective thermal resistance between B **and** $D = 2R$



Temperature of interface $\theta = \frac{R_1\theta_2 + R_2\theta_1}{R_1+R_2}$

$\theta = \frac{R\times 20 + 2R\times 200}{R+2R} = \frac{420}{3} = 140^{\circ}\mathrm{C}$

Q.6 Answer:

$200\ \mathrm{s}$

Solution:

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$\frac{dT}{dt} = \frac{eA\sigma}{mc}\left(T^4 - T_0^4\right) = \frac{e\left(6a^2\right)\sigma}{\left(a^3 \times \rho\right)c}\left(T^4 - T_0^4\right)$

$\Rightarrow$ **For the same fall in temperature, time** $dt \propto a$

$\frac{dt_2}{dt_1} = \frac{a_2}{a_1} = \frac{2\text{ cm}}{1\text{ cm}} \Rightarrow dt_2 = 2 \times dt_1 = 2 \times 100\text{ s} = 200\text{ s}$

(As $A = 6a_2$ **and** $m = V \times \rho = a_3 \times \rho$**)**

Q.7 Answer:

$\left(\frac{\pi}{6}\right)^{1/3} : 1$

Solution:

$Q = \sigma A t\left(T^4 - T_0^4\right)$

If T, T_0, σ **and** t **are same for both bodies then**

But according to problem, volume of sphere = volume of cube

$\Rightarrow \frac{4}{3}\pi r^3 = a^3$

$\Rightarrow a = \left(\frac{4}{3}\pi\right)^n r$

Substituting the value of a **in Eq. (i), we get**

$$\frac{Q_{\text{sphere}}}{Q_{\text{cate}}} = \frac{4\pi r^2}{6a^2} \quad \ldots(\text{i})$$

$$= \frac{4\pi r^2}{6\left\{\left(\frac{4}{3}\pi\right)^{1/3} r\right\}^2} = \frac{4\pi r^2}{6\left(\frac{4}{3}\pi\right)^{23} r^2}$$

$$= \left(\frac{\pi}{6}\right)^{1/3} : 1$$

Q.8 Answer:

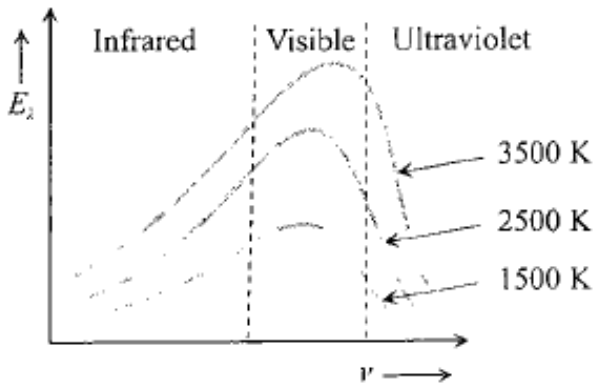
$(3/2)T$

Solution:

Temperature of mixture is given by

$T = \frac{m_1c_1T_1 + m_2c_2T_2}{m_1c_1 + m_2c_2} = \frac{mc2T + \frac{m}{2}2cT}{mc + \frac{m}{2}2c} = \frac{3}{2}T$

Q.9 Answer:



Solution:

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As the temperature of body increases, frequency corresponding to maximum energy in radiation (v_m)) increases. This is shown in graph (c).

Q.10 Answer:

$(C+\gamma_C-S)/3$

Solution:

Apparent coefficient of volume expansion for liquid

$\gamma_{app} = \gamma_L - \gamma_t$

$\therefore \gamma_L = \gamma_{\text{wp}} + \gamma_1$

where γ_s is coefficient of volume expansion for solid vessel.

When liquid is placed in copper vessel

$\gamma_L = C + \gamma_{\text{copper}}$

(As $\gamma_{\text{app.}}$ for liquid in copper vessel $= C$)

When liquid is placed in silver vessel\}

$\gamma_L = S + \gamma_{\text{silver}}$

(As γ_{rp} for liquid in silver vessel $= S$)

From Eqs. (i) and (ii), we get $C + \gamma_{\text{coper}} = S + \gamma_{\text{silver}}$

$\therefore \quad \gamma_{\text{silver}} = C + \gamma_{\text{copper}} - S$

Coefficient of volume expansion $= 3\times$ Coefficient of linear expansion

$\Rightarrow \alpha_{\text{silver}} = \frac{\gamma_{\text{silver}}}{3} = \frac{C+\gamma_{\text{copper}}-S}{3}$

Q.11 Answer:

x, r and d all increase

Solution:

On heating the system; x, r and d all increase, since the expansion of isotropic solids is similar to true photographic enlargement.

Q.12 Answer:

$\frac{m_1c_1T_1+m_2c_2T_2+m_3c_3T_3}{m_1c_1+m_2c_2+m_3c_3}$

Solution:

Let the final temperature be $T^\circ\text{C}$.

Total heat supplied by the three liquids in coming down to 0°C

$= m_1c_1T_1 + m_2c_2T_2 + m_3c_3T_3$ **...(i)**

Total heat used by three liquids in raising temperature from 0°C to $T^\circ\text{C}$

$= m_1c_1T + m_2c_2T + m_3c_3T$ **...(ii)**

By equating Eqs. (i) and (ii) we get

$(m_1c_1 + m_2c_2 + m_3c_3)\,T = m_1c_1T_1 + m_2c_2T_2 + m_3c_3T_3$

$\Rightarrow T = \frac{m_1c_1T_1+m_2c_2T_2+m_1c_3T_3}{m_1c_1+m_3c_1+m_1c_1}$

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Q.13 Answer:

Slope of line AB is $5/9$

Solution:

Relation between Celsius and Fahrenheit scale of temperature is $\frac{C}{5} = \frac{F-32}{9}$

By rearranging we get, $C = \frac{5}{9}F - \frac{160}{9}$

By equating above equation with standard equation of line $y = mx + c$ **we get** $m = \frac{5}{9}$

Q.14 Answer:

$8 : 1$

Solution:

Let the radii of the spheres be $R, R + a, R + 2a$ **and** $R+ 3a$**, where** a **is a constant and the specific heat capacities be** s, sr, sr^2 **and** sr^3 **where** r **is another constant.**

Given, $\left(\frac{\text{heat capacity of } D}{\text{heat capacity of } B}\right) : \left(\frac{\text{heat capacity of } C}{\text{heat capacity of } A}\right) = 8 : 27$

or, $\left[\frac{(R+3a)^3 sr^3}{(R+a)^3 sr}\right] : \left[\frac{(R+2a)^3 sr^2}{Rs}\right] = 8 : 27$

or, $\left(1 + \frac{2a}{R+a}\right) : \left(1 + \frac{2a}{R}\right) = 2 : 3$

or, $\frac{2a}{R+a} : \frac{2a}{R} = 1 : 2$

or, $R = a$

$\frac{m_2}{m_1} = \frac{(4/3)\pi(R+R)^5\rho}{(4/3)\pi(R)^3\rho} = \frac{8}{1}$

Q.15 Answer:

9 min

Solution:

According to Newton's law of cooling

$\left(\frac{\theta_1-\theta_2}{t}\right) = K\left[\left(\frac{\theta_1+\theta_2}{2} - \theta_0\right)\right]$

So that $\left(\frac{80-50}{5}\right) = K\left[\left(\frac{80+50}{2} - 20\right)\right]$

And $\left(\frac{60-30}{t}\right) = K\left[\left(\frac{60+30}{2}\right) - 20\right]$

Solving these for t**, we get** $t = 9$ min.

Q.16 Answer:

More dangerous

Solution:

Work done in converting 1 g **of ice at** -10°C **to steam at** 100°C

$=$ **Heat supplied to raise temperature of** 1 g **of ice from** -10°C **to** 0°C $(m \times c_{\text{ise}} \times \Delta T)$

$+$ **Heat supplied to convert** 1 g **ice into water at** 0°C$(m \times L_{\text{ice}}) +$ **Heat supplied to raise temper.** 0°C **to** 100°C $(m \times c_{\text{water}} \times \Delta T)$

$+$ **Heat supplied to convert** 1 g **water into steam at** 100°C $[m \times L_{\text{vapour}}]$

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$= [m \times c_{\text{ice}} \times \Delta T] + [m \times L_{\text{ice}}] + [m \times c_{\text{water}} \times \Delta T] + [m \times L_{\text{vapour}}]$
$= [1 \times 0.5 \times 10] + [1 \times 80] + [1 \times 1 \times 100] + [1 \times 540] =$
$725\text{cal} = 725 \times 4.2 = 3045 \text{ J}$

Q.17 Answer:

$256/81$

Solution:

According to Wien's law, wavelength corresponding to maximum energy decreases. When the temperature of black body increases, i.e., $\lambda_m T =$ constant
$\Rightarrow \frac{T_2}{T_1} = \frac{\lambda_1}{\lambda_2} = \frac{\lambda_0}{3\lambda_0/4} = \frac{4}{3}$
Now according to Stefan's law
$\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4 = \left(\frac{4}{3}\right)^4 = \frac{256}{81}$

Q.18 Answer:

$P'T' = 32PT$

Solution:

For a black body, wavelength for maximum intensity:
$\lambda \propto \frac{1}{T} \& P \propto T^4 \Rightarrow P \propto \frac{1}{\lambda^4}$
$P' = 16P \quad \therefore P'\text{T}' = 32PT$

Q.19 Answer:

$Q_4 > Q_3 > Q_2 > Q_1$

Solution:

$Q_4 = mL_n = 540\text{cal}$
$Q_3 = ms_w(100 - 0) = 100\text{cal}$
$Q_2 = mL_{\text{ke}} = 80\text{cal}$
$Q_1 = ms_{\text{ven}}(20 - 0) = 20\text{cal}$
$Q_4 > Q_3 > Q_2 > Q_1$

Q.20 Answer:

82 cal

Solution:

Heat required to raise the temperature of m grams of substance by dT is given as
$dQ = mcdT \Rightarrow Q = \int mcdT$

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Therefore, to raise the temperature of 2 g of substance from $5^{\circ}\mathrm{C}$ to $15^{\circ}\mathrm{C}$

$$Q=\int_{5}^{15} 2\times\left(0.2+0.14t+0.023t^{2}\right)dT$$

$$=2\times\left[0.2t+\frac{0.14t^{2}}{2}+\frac{0.023t^{3}}{3}\right]_{5}^{15}=82\text{cal}$$

Q.21 Answer:

—

Solution:

Rate of conduction $R\propto\frac{r^{2}}{\ell}$

The ratio of conduction in them is

$\frac{r^{2}}{\ell}:\frac{r^{2}}{2\ell}:\frac{2r^{2}}{\ell}:\frac{4r^{2}}{\ell}$, i.e. $2:1:4:8$

So, the ratio of maximum to minimum conduction rate is $8:1$

Q.22 Answer:

—

Solution:

$\Delta L=L\times\alpha\times\Delta T=L\times\frac{\gamma}{3}\times\Delta T$

$\left(\text{As }\alpha=\frac{\gamma}{3}\right)$

$\Rightarrow 2\pi\triangle R=2\pi R\left(\frac{\gamma}{3}\right)\Delta T$

$\Rightarrow\Delta T=\frac{3}{\gamma}\frac{\Delta R}{R}=\frac{3\times3}{3.6\times10^{-5}\times497}$

$[\text{As }\Delta R=3\text{ mm and }R=497\text{ mm}]$

Q.23 Answer:

—

Solution:

Let power lost to surrounding is Q

$16-Q=\left(\frac{dm}{dt}\right)S(10)$

and $\quad 32-Q=3\left[\left(\frac{dm}{dt}\right)S(10)\right]$

$\Rightarrow\quad\frac{32-Q}{16-Q}=3\Rightarrow Q=8\text{ W}$

Q.24 Answer:

—

Solution:

From $\frac{(\theta_1-\theta_2)}{t}=\alpha\left(\frac{\theta_1+\theta_2}{2}-\theta_0\right)$, in the first case we have,

$\frac{75-65}{2}=\alpha\left[\frac{75+65}{2}-30\right]$ **...(i)**

For the second case,

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$\frac{55-45}{t} = \alpha\left[\frac{55+45}{2} - 30\right]$ **...(ii)**

Divide Eq. (i) by Eq. (ii), $\frac{t}{2} = 2 \Rightarrow t = 4 \text{ min}$

Q.25 Answer:

—

Solution:

$$\frac{\text{Temperature difference}}{\text{Thermal resistance}} = L\left(\frac{dm}{dt}\right)$$

$$\frac{dm}{dt} \propto \frac{1}{\text{Thermal resistance}}$$

$$q \propto \frac{1}{R}$$

The rods are in parallel in the first case and they are in series in the second case.

$\frac{q_1}{q_2} = \frac{2R}{(R/2)} = 4$

Q.26 Answer:

—

Solution:

Because the sliding plug stays in the connecting pipe, the pressure in both the vessels at the level of the pipe must be the same.

$h_{20}d_{20} = h_{80}d_{80} \Rightarrow \frac{h_{20}}{h_{80}} = \frac{d_{80}}{d_{20}}$

$\Rightarrow \frac{d_0(1-80\gamma)}{d_0(1-20\gamma)} = 0.94$

Q.27 Answer:

—

Solution:

Energy released by water from 25°C **to** $0^\circ\text{C} = 2500 \times 1 \times 25 = 62500\text{cal}$

Energy to bring ice 0^0C

$= 2000 \times \frac{1}{2} \times 15 = 15000\text{cal}$

Energy used to melt ice of m **gram** $= \text{m}80\text{cal}$ **.** $\therefore$ **Ice melt** $\text{m} = \left(\frac{62500-15000}{80}\right) = 593.75\ \text{g}$

So, mass of water $= (2500 + 593.75)\text{g} = 3093.75 \simeq 3\ \text{kg}$

Q.28 Answer:

—

Solution:

Energy with $5\ \text{kg}$ **of** H_2O **at** 20°C **to become ice at** 0°C

$E_1 = 5000 \times 1 \times 20 = 100000\text{cal}$

Energy to raise the temperature of $2\ \text{kg}$ **ice from** -20°C **to** 0°C

$E_1 = 5000 \times 0.5 \times 20 = 20000\text{cal}$

$(E_1 - E_2) = 80000\text{cal}$ **is available to melt ice at** 0°C**. So only** $1000\ \text{g}$ **or** $1\ \text{kg}$ **of ice would have**

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melt.

So, the amount of water available $1 + 5 = 6$ kg.

Q.29 Answer:

—

Solution:

Loss or gain per day $= dT = \frac{1}{2}\alpha dt \times 86400$

Since $T = 86400$ s **for each day**

At $15^{\circ}\mathrm{C}, \quad 5 = \frac{1}{2}\alpha(t - 15) \times 86400$

Q.30 Answer:

—

Solution:

The brass rod and the lead rod will suffer expansion when placed in steam bath.

$\therefore$ **Length of brass rod at** $100^{\circ}\mathrm{C}$.

$L_{\text{tram}} = L_{\text{tran}}\ (1 + \alpha_{\text{trm}}\Delta T) = 80\left[1 + 18 \times 10^{-6} \times 100\right]$

and the length of lead rod at $100^{\circ}\mathrm{C}$

$L_{\text{kad}} = L_{\text{cad}}\ (1 + \alpha_{\text{less}}\,\Delta T) = 80\left[1 + 28 \times 10^{-6} \times 100\right]$

Separation of free ends of the rods after heating $= L_{\text{leat}} - L_{\text{sens}} = 80[28 - 18] \times 10^{-4} = 8 \times 10^{-2}$ cm $= 0.8$ mm

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